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EMC2: Energy-efficient and multi-resource- fairness virtual machine consolidation in cloud data centres

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ABSTRACT

The rapid rise in the cloud service adoption reflects the growth of Cloud Data Centers' (CDCs) number, size, energy consumption and eco-unfriendly carbon footprints. In CDCs, Virtual Machine Consolidation (VMC) plays a significant role in reducing their energy consumption and thereby reducing the carbon footprints. The state-of-the-art VMC heuristics based on First-Fit Decreasing (FFD) and Dominant Residual Resource (DRR) called DRR-FFD and DRR-BinFill are grouping the VMs based on single VM resource. We attempt to further reduce the energy consumption of CDCs through the proposed EMC2, an energy-efficient VMC framework that employs our multi-resource-fairness based VM selection heuristics, namely VMNeAR-H (Hierarchical), VM NeAR- D (Directed Hierarchical) and VM NeAR-E (Euclidean Distance). A dataset extracted from ENERGY STAR® containing the heterogeneous physical machine resource capacities and their estimated energy consumptions is utilised in the simulation experiments. The proposed EMC2-VMNeAR-D heuristic dominates the existing DRR heuristics in terms of total energy consumed by all the physical machines in the CDC (3.318 % energy savings on average of 40 schedules = 185107 kWh).

1. Introduction

The unprecedented growth in the adoption of contemporary Information Technology services like cloud computing, big-data and internet-of-things reflect the rise of Cloud Data Centers (CDCs), both in number and size. The similar surge in their energy usage impacts the environment adversely. Efficient energy management in the CDCs is a crucial concern of immediate focus to save energy, reduce operational costs and harm to the environment. The energy-related costs are estimated to be 42 % of the total operational cost of CDCs [1]. On-demand resource provisioning may also lead to SLA violations and attracts penalty [2]. Thus a CDC management is a complex activity supported by several layers of automated management systems like Virtual Machine Consolidation (VMC). VMC uses bin-packing heuristics to consolidate Virtual Machines (VMs) into Physical Machines (PMs). The current problem, placement of VMs into heterogeneous PMs is a variable size vector bin-packing problem and is discussed in the sub-section1.3.

1.1. Bin-packing problem (BPP)

The goal of the standard bin-packing optimisation problem is to pack the given set of n items $I = \{I_1, \ldots, I_n\}$, into a minimum number of same capacity bins. A solution is said to be feasible if it satisfies the following two conditions: (i) each item is to be placed in one and only one bin (ii) the sum of sizes of all the items packed in a bin should not exceed the bin capacity. The problem is NP-Hard, and the approximate algorithms like First-Fit Decreasing (FFD), Best-Fit Decreasing (BFD) are widely in use.

1.2. Multi-capacity bin-packing problem

In this generalised bin-packing problem, the items and bins have multiple capacities. However, the goal and constraints of this optimisation problem are similar to the standard problem, BPP. VMC is a multi-capacity bin-packing problem. Both VM and PM have multiple resource demands and capacities, respectively. Each VM treated like an item and PM as a bin. A feasible solution satisfies the following two conditions: (i) Each VM is to be placed in one and only one PM. (ii) For each resource capacity of a PM, the sum of current resource demand by all the VMs placed in that PM should not be exceeded. FFD variants are usually used to solve this NP-Hard, as well as the APX-Hard problem. The problem is also known as Vector Bin-Packing (VBP).

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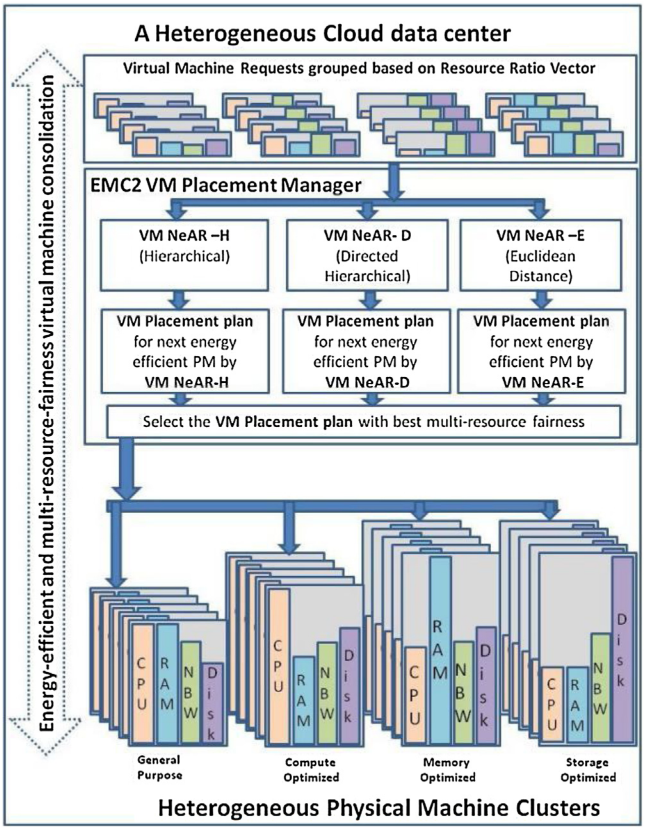

Fig. 1. The architecture of the Cloud Data Center used in simulation experiments.

1.3. Variable size vector bin-packing problem

VSVBP is another generalisation of BPP, and it deals with bins of different cost and capacities. The objective of this optimisation problem is to pack the given items in a set of bins with minimum total cost.

The current research work proposes novel heuristics for VSVBP in a CDC for Energy-Efficient (EE) VM consolidation. The architecture model of the CDC used in the simulation experiments is depicted in Fig. 1. The CDC is hosting PMs with different resource capacities designed to cater to the user's resource demands like General Purpose, Compute Optimised, Memory Optimised and Storage Optimized, inspired by Amazon EC2.[1]

FFD is a polynomial-time approximation scheme (PTAS) for one-dimensional or standard BPP. The first-fit heuristic places each item in the first bin it fits among the opened bins, opens a new bin if the item does not fit in any of the opened bins. FFD supplies the items to the first-fit heuristic in decreasing order. The reason is to fill the more significant items first and fill the gaps with smaller items in the last. FFD is extended for VBP by introducing vector to scalar conversion methods. FFD based approaches can be applied merely if all the PMs in the CDC are homogenous. Later, Panigrahy et al. [3] modified item-filling style of FFD to bin-filling variant and made it suitable for filling VMs in heterogeneous PMs too. Microsoft's virtual machine manager internally uses First-Fit Decreasing (FFD) oriented bin-packing techniques called dot-product and norm-based greedy for VMC [4]. The current research work is focused on Variable Size Vector Bin-Packing (VSVBP). The objective of VSVBP is to minimise the total cost of bins used to pack the given items. The energy-efficient VM consolidation in a CDC with heterogeneous PMs is modelled as a VSVBP to pick the set of PMs that consumes the minimum energy in total for hosting the VM requests.

1.4. Heterogeneity aware dominant resource assistant heuristics

The heterogeneity aware Dominant Resource Assistant (DRA) heuristics proposed by Zhang et al. [5] are most appropriate to compare with the current research work focusing on Multi-Resource-Fairness (MRF). DRA divides the VMs into a maximum of d groups, each for a resource when resources are considered in VMC problem. Like FFD, DRA also places the largest item every time, but from the VM group with the same dominant resource as PM's residual resources.

Our proposed EMC2 framework use VM NeAR heuristics for choosing the next VM to be placed based on PM's residual resource ratios. VM NeAR heuristics assure MRF, fills faster than FFD and DRA heuristics, the time complexity analysis is discussed in Section 4.2. The rest of the paper is organised as follows: Section 2 defines the VMC problem and discusses the related work. The proposed EMC2 framework, VM NeAR and MRF operators are detailed in Section 3. Section 4 presents and discusses the empirical results. Finally, Section 5 concludes the article along with the future scope of extension.

2. Virtual machine consolidation in CDCs hosting heterogeneous PMs

The existing literature solved VMC using heuristics, hyper-heuristics, meta-heuristics, machine learning, multi-agent systems and other hybrid optimisation techniques. However, most of the contributions are focused on the VMC problem dealing with homogenous PMs. All the PMs hosted in a CDC need not be with the same resource capacities and energy consumptions. The rise in demand for cloud service adoption is leading to an expansion in the CDCs, possibly with the new generation PMs that have low carbon emission and green computing facilities. ENERGY STAR®[2] is analysing and listing the energy consumption information of all the electrical appliances, including computing devices. Table 1 is the extract of a type of PMs called Blade servers from the ENERGY STAR® product finder listing.

2.1. Problem formulation

The current research problem, VMC in a CDC hosting heterogeneous PMs is formulated as a VSVBP. The focus of VSVBP is to optimise the total cost of the subset of heterogeneous bins (PMs) chosen to pack the items (VMs). The optimisation objective is to place the given VMs onto a subset of PMs with total estimated energy consumption is low. The energy consumption of a PM has two parts: fixed and variable parts. Even when idle, the PM consumes the fixed part around 60 % of it's consumption during peak load. The variable part depends on the current load. Thus the PMs Energy consumption can be formulated as in Eq. (1).

$$E(Current\ Load) = F + V(Current\ Load) \tag{1}$$

The energy consumptions of PMs listed in Table 1 is profiled under max-power configuration with peak load. To identify the energy-efficient PMs, we are applying a metric: Energy consumption per CPU Core (EpC) and is listed in Table 1. Even though the energy consumption of a PM is variable and depends on the load, for a particular load, it is obvious that a PM with less EpC will consume less energy compared to the PM with high EpC. The peak-load

[1] https://aws.amazon.com/ec2/.

[2] https://www.energystar.gov.

Table 1
ENERGY STAR® certified blade servers.

ENERGY STAR®UniqueID	RAM (GB)	CPU Cores	Estimated Energy Consumption on Peak Load(*kWh*)	Energy per CPUCore (*EpC*)
2217079	384	20	706.76	35.34
2233386	1536	40	421.31	10.53
2252850	3072	72	1013.16	14.07
2224984	3072	60	644.48	10.74
2224985	1536	30	318.01	10.60
2213866	768	20	865.10	43.26
2239193	768	24	248.15	10.34
2211013	768	40	165.73	4.14
2234713	512	64	275.60	4.31
2247547	256	28	165.20	5.90
2225991	256	28	283.85	10.14
2234797	512	72	596.60	8.29
2229224	64	32	59.65	1.86
2204277	192	16	94.79	5.92
2203918	192	16	137.27	8.58
2204276	384	16	322.15	20.13
2204692	768	16	335.20	20.95
2204703	768	16	154.10	9.63
2226290	256	28	264.50	9.45
2226541	256	28	158.16	5.65
2231915	256	28	264.50	9.45
2232402	256	28	264.50	9.45
2203920	768	32	291.95	9.12
2234828	512	72	299.28	4.16
2203503	768	24	282.89	11.79
2231810	512	32	2499.60	78.11
2254470	1024	72	1235.28	17.16
2257723	1536	36	289.68	8.05
2216144	32	32	1622.00	50.69
2238248	128	24	120.00	5.00
2233219	1536	28	113.30	4.05
2272759	512	28	139.16	4.97
2238256	384	24	175.48	7.31
2248297	1536	28	128.84	4.60
2238255	384	32	185.70	5.80
2206615	32768	12	241.21	20.10
2238240	32768	16	303.04	18.94

energy consumptions are used to calculate the estimated energy of the solutions.

Let the matrix $P^{Nx(d+2)}$ holds the N types of PMs, each with d resource capacities, its estimated energy consumption and a number of such PMs available. Each VM has d resource demands (I^d) and all the n VMs are represented as a list of vectors L. $F(x, y)$ be a function to denote whether VM x is placed in PM y or not, the value of the function is 1 or 0 respectively. $R(i)$ be a function to indicate the running state of the PM_i, 0 if idle and 1 otherwise.

The problem is to assign all the given n VMs on to PMs such that:

$$Minimize \sum_{i \in P} E_i(Full\,Load)^* R(i) \tag{2}$$

$$\sum_{i \in P} F(i, j) = 1\ ,\ \forall j \in L \tag{3}$$

$$\sum_{j \in L} F(i, j)^* I_j^r \leq P_i^r\ ,\ \forall i \in P \tag{4}$$

Eq. (2) is the optimisation objective; the constraint in Eq. (3) ensures that each VM is assigned to one and only one PM. Eq. (4) governs the multiple resource capacity constraints.

2.2. Related work

VMC problem belongs to NP-Hard class of problems. There is no exact algorithm that produces an optimal solution. The state-of-the-art review on VMC approaches is presented in Table 2, approximate algorithms like FFD, several other greedy heuristics, meta-heuristics and machine learning approaches, have been proposed to minimise the energy consumption and resource wastage. The simulation experiments were carried out using VM instances inspired by Amazon EC2, synthetic data, random and real workload traces.

Inspired by the bin-packing heuristics, a modified best fit decreasing (MBFD) algorithm was proposed by Beloglazov et al. in [7], through a continuous search of each VM and PM combination to find the minimum pow- er-consuming combinations on the whole to reduce the CDC energy consumption. Xu and Fortes [9] proposed a multi-objective Grouping Genetic Algorithm (GGA) for VMC. GGAs are used to explore the permutations of the given VMs as a search procedure to find the best permutations of VMs, which consumes less energy when placed on PMs. The objective of the algorithm is to minimise the power consumption and resource wastage by searching the solution space. VMPACS [10], a VM placement based on ant colony system optimisation outperforms the GGA in [9]. The focus of VMPACS is to reduce the resource wastage and thus energy consumption in a CDC. VMPACS consumes extra energy during placement optimisation in the form of migrations. Particle Swarm Optimization (PSO) was used in [11] to minimise energy consumption. The PSO algorithm is modified to update the particle positions with energy-aware local fitness first strategy. An Artificial Bee Colony (ABC) algorithm is proposed by Jiang et al. in [12] for EE practices in CDCs using live VM migration. Another migration based method in [19] also improves system reliability along with energy minimisation. The VMC approach by Li et al. in [20] using multi-dimensional-space-partition-model focusing on balanced multiple resource utilisation. To minimise the resource wastage and energy consumption, Gupta and Amgoth developed

Table 2
State of the art Review of VM consolidation approaches.

VMC Approach	Authors	Algorithm(s)	Focused on minimising	VM Instance Dataset
VBP heuristics	Panigrahy et al., 2011	FFDSum, FFDProd, Dot Product, Norm-based greedy [3]	PMs used	Random instances
	Mishra et al., 2011	Static VM Placement [6]	PMs used	No details
	Beloglazov et al., 2012	Modified best fit decreasing (MBFD) [7]	Energy consumption and SLA violation	Random CPU utilisations
	Zhang & Ansari, 2013	DRR-BinFill [5]	Total estimated energy consumption by PMs	Random instances
	Canali et al., 2017	Class-based placement [8]	PMs used	VM traces of real applications
Meta-heuristics	Xu and Fortes, 2010	Grouping Genetic Algorithm (GGA) [9]	Total resource wastage, power consumption and thermal dissipation costs	Synthetic Data
	Gao et al., 2013	VMPACS [10], a VM placement based on ant colony system optimisation	Power consumption and resource wastage	Random instances
	Wang et al., 2013	Particle Swarm Optimi- zation (PSO) [11]	Energy consumption	Amazon EC2: Micro Instance and Small Instance
	Jiang et al., 2017	Artificial Bee Colony (ABC) algorithm [12]	Total estimated energy consumption by PMs	PlanetLab Data
	Satpathy et al., 2017	Crow search based virtual machine placement [13]	Power consumption and resource wastage	Inspired by Amazon EC2
Machine Learning Approaches	Masoumzadeh et al., 2015	DDVMC [14]	Energy consumption	Real workloads
	Masoumzadeh et al., 2015	DFQL Host Overload-ing Detection [15]	Energy consumption and SLA violation	PlanetLab Data
	Tang et al., 2016	CBVMP [16], classification based VM Placement	Resource wastage	Random instances
	Jobava et al., 2018	Enhanced OMA [17]	Communication cost	Random traffic data
	Ranjbari et al., 2018	Learning automata overload detection (LAOD) [18]	Energy consumption and SLA violation	PlanetLab Data

a VMC method in [21] for a CDC which hosts homogeneous PMs that exploit the concept of harmonic mean. There is no claim of the heterogeneous environment of PMs for VMC in all the above approaches, and testing is also done only in homogenous environments. The proposed EMC2 framework is for both the homogenous and heterogeneous cases.

The other works focused on a CDC with heterogeneous PMs are discussed here: A column generation approach was proposed by Kramer et al. in [22] focusing on CPU power-saving techniques for optimisation of energy consumption in heterogeneous PMs. Park et al. [23] conducted mathematical stability analysis to balance the load levels across the heterogeneous PMs and stabilise the system load at a steady state. Lent [24] evaluated the cooling and computing energy demands for optimising the cooling energy. CPU dynamic voltage frequency scaling methods were proposed in [25] focus on heterogeneous servers. Game theory was exploited in [26] by Khani et al. for EE dynamic VMC in heterogeneous CDC. A genetic algorithm EEE-VMA is proposed in [27] for optimal placement of collocated VMs. The idleness of a VM is predicted and exploited for EE practice in [28] by migration based Smart VM Over Provision. Two greedy approximation algorithms were proposed by Dai et al. [29] for minimising network energy consumption by placing the communicating VMs in the same PM or the same rack. In [30], a VMC approach based on biogeography-based optimisation focus to minimise both the resource wastage and energy usage. This work also highlights the multiple resource (CPU, RAM, Disk Storage and Bandwidth) wastage minimisations. A VM migration based method using Chicken Swarm Optimization is proposed in [31] to carry out dead-lock free migrations in heterogeneous servers. Kharghariaet *al.* used game theory to optimise performance per watt at the cluster level of a data center. The power consumption reduces by 65 % when compared to traditional heuristic techniques [32]. Our hybrid heuristics in [33] are proved to be efficient in reducing the homogenous physical machines used, and the migration management of virtual machines is addressed in [34]. The role of

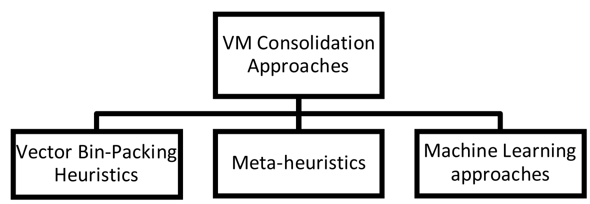

Fig. 2. Three significant Categories of VM Consolidation approaches.

elastic computing infrastructure for efficient energy management of cloud data centres is discussed in [35]. The initial placement of virtual machines in the absence of power consumption is addressed through the vector bin-packing heuristic PMNeAR-vector [36].

On the whole, the VMC approaches in the literature can be categorised as follows: Vector Bin-packing heuristics, Meta-heuristics and Machine Learning techniques as shown in Fig. 2. Most of the existing works based on Machine Learning and Meta-heuristics are focused on VM migration. VM migration based methods are a better option for dynamic consolidation. The dependency of a VMC approach for initial placement on VM migration leads to additional energy consumption. Our current EMC2 is a vector bin-packing based method of initial placement and is free from migrations.

3. Proposed EMC2 framework

Energy-efficient and Multi-resource-fairness Consolidation in Cloud (EMC2) framework is proposed using (Energy per CPU Core) as a metric to choose the EE PM, and Virtual Machine Nearest and Available to Residual resource ratios of PM (VM NeAR) [37] as multi-resource fairness VM selection operator. Three such operators called VMNeAR-H, VMNeAR-D and VMNEAR-E are introduced in the proposed EMC2 framework.

Table 3
Description of MRF Operators.

MRF Operator	Description of nearest VM group computation
VM NeAR–H (Hierarchical)	Each resource ratio is compared one by one hierarchically.
VM NeAR-D (Directed Hierarchical)	Hierarchical is extended in this direction (+,-) aware operator. If a PM residual resource ratio is less Compare to the PM's initial resource ratios, the direction will be (+) indicating a higher resource ratio is to be chosen and similarly for (-).
VM NeAR-E (Euclidean Distance)	Based on the Euclidean distance from the PM residual resource ratio vector, the VM group with less distance is selected.

3.1. Resource ratios and resource ratio vector

The EMC2 framework uses VMNeAR (H/D/E) to determine the multi-resource-fair VM group. VMNeAR uses current PMs residual resource ratios and the available VMs resource ratios to determine so. Each VM with d resource demands is represented using a resource ratio vector of length $d-1$. All the remaining $d-1$ resource demands are represented in proportion to CPU cores. CPU Cores is treated as size to differentiate VMs with the same resource ratios. A VM request with four resource demands namely CPU Cores, RAM, Bandwidth and Disk Storage is represented using $\left(CPU\,Cores,\ \left\langle \frac{RAM}{CPU\,Cores}, \frac{Bandwidth}{CPU\,Cores}, \frac{Disk\,Storage}{CPU\,Cores} \right\rangle\right)$ as $(Size, Resource\,ratio\,vector)$. Similarly, the current PM is represented with residual resource ratios using remaining resource capacities upon placing each VM.

Algorithm 1: EMC² framework using VMNeAR

1. Group VMs based on resource ratio vector
2. Identify the least resource demand for the resources
3. Arrange PMs in the ascending order of EpC
4. While (all the VMs are not placed)
5. Start the next idle PM with low EpC
6. While (residual resource(r_i) > least resource demand(l_i), $\forall i = 1\ to\ d$)
7. Identify the best multi-resource-fair VM group usingVMNeAR
8. Place the largest VM (in the group) that fits, if none fits, go to step 5
9. End While
10. End While

The working of EMC2 framework in Algorithm 1 is as follows: In step 1, the VMs are grouped based on the resource ratios. Step 2 identifies the least demand for each resource among all the VM requests and is used in step 6 to determine whether any more VM fits well in the current PM. Steps 3 and 5 support energy efficiency by starting the PMs in the increasing order of their EpC. Step 4 ensures that all the VMs are placed. In Step 7, one of the VMNeAR (H/D/E) identifies the best Multi-Resource-Fairness (MRF) group of VMs, the working is explained in a case study below. Step 8 places the possible largest VM of that group that fits in the current PM; if no VM fits, the VM placement process continues to the next PM.

3.2. VM NeAR-(H/D/E)

VM Nearest and Available to Residual resource ratios of PM (VM NeAR) is the Multi-Resource-Fairness VM group identification heuristic according to the current PM residual resource ratios. Initially, the VMs with the same resource ratios are grouped based on resource ratio vectors. The working of all three MRF operators is described in Table 3 and illustrated with a case study.

3.2.1. Case study

Consider there are VM groups with resource ratios $\langle 1,2,3,4\rangle$, $\langle 2,1,3,4\rangle$, $\langle 1,2,4,3\rangle$, $\langle 4,3,2,1\rangle$ and the PM's initial and current residual resource ratios are $\langle 1.5,2,3,4\rangle$ and $\langle 1.4,2.3,3.2,3.5\rangle$ respectively.

3.2.2. VMNeAR-Hierarchical (H)

For the PM's first residual resource ratio 1.4, 1 is nearest among the VM's first resource ratios (1, 2, 4). Now, there are two VM groups with 1 as the first resource ratio $\langle 1,2,3,4\rangle$ and $\langle 1,2,4,3\rangle$. The same is the case in the second resource ratio too. The third resource ratio 3.2 is near to $\langle 1,2,3,4\rangle$ than $\langle 1,2,4,3\rangle$. Hence $\langle 1,2,3,4\rangle$ is the nearest MRF VM group according to VMNeAR-H.

3.2.3. VMNeAR-Directed hierarchical (D)

Here direction vector is used, i.e., $\langle +,-,-,+\rangle$ from the difference between initial and current residual resource ratios of PM $\langle 1.5-1.4, 2-2.3, 3-3.2, 4-3.5\rangle$. Now the first residual resource ratio 1.4 is near to 1 compared to the other 2 and 4, but the direction + indicates to choose the nearest value above 1.4, i.e., 2. Hence $\langle 2,1,3,4\rangle$ is the nearest MRF VM group according to VMNeAR-D.

3.2.4. VMNeAR-Euclidean distance (E)

The Euclidean distances from the VM groups $\langle 1,2,3,4\rangle$, $\langle 2,1,3,4\rangle$, $\langle 1,2,4,3\rangle$ and $\langle 4,3,2,1\rangle$ to PMs residual resource ratios $\langle 1.4,2.3,3.2,3.5\rangle$ are 0.734847, 1.529706, 1.067708, 3.86523. Hence $\langle 1,2,3,4\rangle$ is the nearest MRF VM group according to VMNeAR-E.

4. Results and discussion

4.1. Algorithms and data used for comparison

The proposed EMC2 framework with MRF operators VMNeAR(H/D/E) is compared with existing FFD and DRF based algorithms FFDProd [3] and DRR-BinFill [5] along with our earlier work FFD-Aggregated Rank (FFD-AR) [38]. The implementations were carried out in an Intel® Core ™ i7 processor @ 2.5 GHz system with 4 GB RAM running Windows 10 operating system. All the algorithms were implemented in Python environment. The PMs data is inspired by the data shown in Table 1, and the VMs with different resource demands inspired by Amazon EC2 VM instances. Both PMs and VMs have four resources, namely CPU Cores, RAM, Bandwidth and Disk Storage with three resource ratios and were generated randomly from the experiment running for 40 schedules.

In each schedule, PMs and VMs of different sizes were generated and are supplied the same for all the six VMC methods used for comparison. The estimated energy consumption of all the VMC

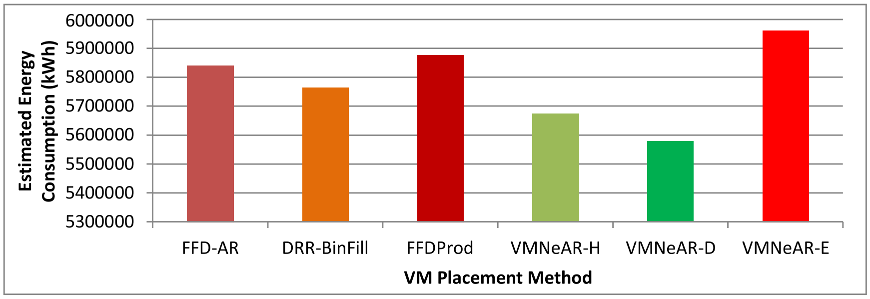

Fig. 3. Estimated total energy consumption of the CDC in 40 schedules by different VM placement methods.

Table 4
Time complexities of VMC methods.

VMC Method	Time Complexity
FFD based	$O(n \log n)$
DRR-BinFill	$O(n)$
FFD-AR	$O(d\, n \log n)$
EMC2	$O(n)$

methods across 40 schedules is listed in Table 5, and a comparison chart of total energy is in Fig. 3. Even though none of the six heuristic VMC methods is completely dominating each other, on average, the EMC2 framework with VMNeAR-D is consuming 3.3 % less energy compared to the baseline method DRR-BinFill. While these methods attempt to minimise the energy consumption of the computational servers, another new research problem is thermal-aware scheduling discussed in [39,40].

4.2. Time complexity analysis

First-fit decreasing (FFD) sorts all the items initially and places the largest item to smallest item one by one in the first bin that fits among the opened bins. FFD is a one-dimensional heuristic and is extended for multi-capacity packing using some weight measures to convert the vector items to a scalar. FFD based FFDProd does the same and its time complexity is $O(n \log n)$, DRR-BinFill is an extension of FFD, where the items are grouped into at most d dominant resource groups, since it is a bin filling strategy there is no need to search for the first bin it fits and hence the time complexity is $O(n)$. FFD-AR ranks the VMs according to each resource demand, and the sum of all the ranks is used as Aggregated Rank to determine the next largest VM to be placed, the time complexity of FFD-AR is $O(d\, n \log n)$. EMC2 framework is a bin filling mechanism, and the time complexity of EMC2 with any MRF operator VMNeAR (H/D/E) is $O(n)$. The time complexities of VMC methods are listed in Table 4.

5. Conclusion and future work

As VMC plays a vital role in energy efficient (EE) management of CDCs, there is a need for state-of-the-art EE frameworks that ensures MRF and minimises the resource wastage. The MRF property can cut down the operational expenses, improve profits and reduce harm to the environment by lowering the carbon footprints. While DRF based methods are focusing on only one dominant resource for choosing a fair VM, here our MRF operators focus on a resource by resource in a hierarchical fashion. The user can customise the hierarchy according to the resource preciousness of the user. According to the power consumption share of each resource, the literature says that the CPU consumes more power than RAM and the other resources in a PM. We are filling all the resources in proportion to CPU Cores. The first resource ratio is used as RAM per CPU Core since RAM is the next high energy consumer. On average in 40 schedules EMC2 framework with VMNeAR-D MRF operator consumed 3.318 % less energy compared to the existing DRF operator based DRR-BinFill VMC method. As a promising future work, the EMC2 framework can be extended with VM swapping into smaller PMs in case of very low resource occupancy in a huge PM.

Table 5
Estimated energy consumption by VMC methods.

Schedule ID	Estimated Energy Consumption (kWh)					
	FFD-AR	DRR-BinFill	FFDProd	VMNeAR-H	VMNeAR-D	VMNeAR-E
1	93343	112917	101766	71252	72620	83245
2	8785	8124	8766	7803	7803	8785
3	4548	4548	4548	4548	4548	4548
4	11968	12166	11996	11343	11418	11752
5	64626	58198	66055	52202	47024	63697
6	184622	180229	182703	172454	176745	190211
7	49581	55659	54560	55240	55765	63118
8	31794	29221	32446	27867	27867	29733
9	70607	75515	69575	74495	66683	70384
10	34476	34932	34384	32888	32948	34728
11	33987	47107	36641	35326	34510	36223
12	2213	2234	2173	2139	2136	2228
13	19876	18376	19314	18418	18418	21148
14	7808	7994	7802	7890	7810	7862
15	17991	18225	18126	18495	18207	17928
16	24318	24318	24318	24433	24318	24318
17	133728	151967	139029	133610	123875	156119
18	703956	768430	654022	701409	660260	655808
19	42480	42434	42455	42066	42041	42495
20	53023	53023	53023	53023	53023	53023

Table 5 (*Continued*)

Schedule ID	Estimated Energy Consumption (kWh)					
	FFD-AR	DRR-BinFill	FFDProd	VMNeAR-H	VMNeAR-D	VMNeAR-E
21	71599	71540	72299	68524	67958	68977
22	37560	38760	37476	37788	37788	37770
23	536	536	536	536	536	536
24	1497	1497	1497	1497	1497	1497
25	10920	11057	11064	10086	10086	11612
26	39876	40419	42877	41263	38296	39287
27	1779024	1801452	1849703	1778291	1777112	1825854
28	1438	1438	1438	1438	1438	1438
29	11541	11209	11841	10689	10679	13911
30	37204	36323	39258	30168	27636	31806
31	9872	9872	9872	9872	9872	9872
32	13480	13632	13480	13296	13296	13552
33	60813	62675	60483	53091	52960	78096
34	1638412	1414313	1616310	1566797	1534303	1705573
35	12518	12366	12140	11838	11785	11753
36	383796	379679	387379	357520	360261	380354
37	61004	65896	60776	60216	60216	61352
38	2631	8372	5916	3313	3313	13911
39	246	246	246	246	246	246
40	73314	76668	79368	71163	71163	77256
Total Energy (kWh)	5841011	5763567	5877661	5674533	5578460	5962006
Excess Energy (%)	4.707 %	3.318 %	5.364 %	1.722 %	0.000 %	6.875 %

Declaration of Competing Interest

No conflict of interest.

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